

09_lagged_mirror

July 2, 2026

1 09 — Realistic lagged mirror (does the exit edge survive latency?)

Notebook 08 showed that mirroring the sharps' **entries AND exits** is profitable at the *frictionless* ceiling ($\sim +11\%$ per position, positive median, consensus helps). The whole project has taught us one lesson though: **latency destroys copy edges**. This notebook is the decisive stress-test.

What it does: for a sample of point-in-time-qualified positions, it reconstructs the sharp's actual round trip from their trade activity (when they bought, when they sold), then simulates a *realistic* copy: - **Enter** COPY_LAG_HOURS after their buy, at the market price *then*, paying slippage. - **Exit** COPY_LAG_HOURS after their sell, at the market price *then*, paying slippage — or, if they never sold, **hold to resolution** like they did.

The gap between this and the notebook-08 ceiling is the tax you pay for being a step behind. If the lagged return is still clearly positive $\rightarrow$ we have a real, tradeable strategy worth paper-trading. If latency eats it $\rightarrow$ the ceiling was a mirage and we've hit the wall.

```
[1]: import importlib, pmc, time, random
importlib.reload(pmc)
from pmc import CFG, get_leaderboard, get_closed_positions, get_user_trades, \
    price_at
import pandas as pd, numpy as np
print(f"copy lag = {CFG.COPY_LAG_HOURS}h | sample = {CFG.MIRROR_SAMPLE} \
positions | "
      f"slippage = {CFG.SLIPPAGE_TOLERANCE}")
```

copy lag = 6h | sample = 600 positions | slippage = 0.06

1.1 1. Rebuild the point-in-time-qualified positions (cached, fast)

Identical to notebook 08: candidate pool $\rightarrow$ resolved history $\rightarrow$ PIT qualification $\rightarrow$ backer count.

```
[2]: cands = {}
for w in ("ALL", "MONTH"):
    for r in get_leaderboard(window=w, limit=CFG.WF_CANDIDATES):
        cands.setdefault(r["wallet"], r)
rows = []
for wallet in cands:
    for p in get_closed_positions(wallet, max_positions=600):
```

```

        cost = float(p.get("totalBought") or 0)
        if cost <= 0:
            continue
        rows.append({"wallet": wallet, "conditionId": p.get("conditionId"),
        ↪ "outcome": p.get("outcome"),
        ↪ "asset": p.get("asset"), "entry": float(p.get("avgPrice"))
        ↪ or 0),
        ↪ "entry_ts": int(p.get("timestamp") or 0), "endDate": p.
        ↪ get("endDate"),
        ↪ "realizedPnl": float(p.get("realizedPnl") or 0), "cost":
        ↪ cost,
        ↪ "won": 1 if float(p.get("realizedPnl") or 0) > 0 else 0})
h = pd.DataFrame(rows)
h["res_ts"] = (pd.to_datetime(h["endDate"], errors="coerce", utc=True).
    ↪ astype("int64") // 10**9)
h = h[(h["entry_ts"] > 0) & (h["res_ts"] > 0) & (h["res_ts"] > h["entry_ts"]) &
    ↪ (h["entry"] > 0)].copy()
h = h.sort_values("entry_ts").reset_index(drop=True)
qmask = np.zeros(len(h), dtype=bool)
for wallet, idx in h.groupby("wallet").groups.items():
    w = h.loc[idx]; rs = w.sort_values("res_ts")
    rt = rs["res_ts"].values; wc = np.cumsum(rs["won"].values)
    k = np.searchsorted(rt, w["entry_ts"].values, side="left")
    wins = np.where(k > 0, wc[np.clip(k - 1, 0, len(wc) - 1)], 0)
    wr = np.where(k > 0, wins / np.maximum(k, 1), 0.0)
    qmask[np.array(idx)] = (k >= CFG.WF_MIN_TRAILING_TRADES) & (wr >= CFG.
    ↪ WF_MIN_TRAILING_WINRATE)
q = h[qmask].copy()
q["backers"] = q.groupby(["conditionId", "outcome"])["wallet"].
    ↪ transform("nunique")
print(f"{len(q)} PIT-qualified positions")

```

4175 PIT-qualified positions

1.2 2. Reconstruct each sharp's exits from their trade activity

We pull each involved wallet's full BUY/SELL history once, then per (wallet, token) find how much they sold and *when they last sold*. That last-sell time is our exit trigger; if they barely sold, we treat it as held to resolution.

```

[3]: sample = q.sample(n=min(CFG.MIRROR_SAMPLE, len(q)), random_state=42).copy()
wallets = sample["wallet"].unique()
print(f"pulling trade history for {len(wallets)} wallets (cached after first
    ↪ run)...")

# (wallet, asset) -> {buy_size, sell_size, last_sell_ts}
flow = {}

```

```

for j, wal in enumerate(wallets):
    for t in get_user_trades(wal):
        key = (wal, t.get("asset"))
        f = flow.setdefault(key, {"buy": 0.0, "sell": 0.0, "last_sell_ts": 0})
        sz = float(t.get("size") or 0)
        if t.get("side") == "BUY":
            f["buy"] += sz
        elif t.get("side") == "SELL":
            f["sell"] += sz
            f["last_sell_ts"] = max(f["last_sell_ts"], int(t.get("timestamp"))
↳ or 0))
    if (j + 1) % 15 == 0:
        print(f" {j+1}/{len(wallets)} wallets")
print(f"reconstructed flow for {len(flow)} (wallet, token) positions")

```

pulling trade history for 53 wallets (cached after first run)...

```

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15/53 wallets
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30/53 wallets
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[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x9703676286b93c2eca71ca96e8757104519a69c2', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0x9703676286b93c2eca71ca96e8757104519a69c2&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0xe5c8026239919339b988fdb150a7ef4ea196d3e7', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0xe5c8026239919339b988fdb150a7ef4ea196d3e7&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0xba389f76b0119aed07c53c9029852664bd97e406', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0xba389f76b0119aed07c53c9029852664bd97e406&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0xa804390f80019699ab34a282c0df7528fba82a75', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0xa804390f80019699ab34a282c0df7528fba82a75&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x26437896ed9df2f69765edcafe8fdceaab39ae', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0x26437896ed9df2f69765edcafe8fdceaab39ae&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x006cc834cc092684f1b56626e23bedb3835c16ea', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0x006cc834cc092684f1b56626e23bedb3835c16ea&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x1a2f67124f12eaa88be0f6683a95abede555647a', 'type': 'TRADE',
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'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0x1a2f67124f12eaa88be0f6683a95abede555647a&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x8b887c9ad761b851d9e14510ea9aae04a01e8f33', 'type': 'TRADE',
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'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0x8b887c9ad761b851d9e14510ea9aae04a01e8f33&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

45/53 wallets

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x44c1dfe43260c94ed4f1d00de2e1f80fb113ebc1', 'type': 'TRADE',
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'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0x44c1dfe43260c94ed4f1d00de2e1f80fb113ebc1&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x14964aefa2cd7caff7878b3820a690a03c5aa429', 'type': 'TRADE',
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'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0x14964aefa2cd7caff7878b3820a690a03c5aa429&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

[warn] GET failed after retries: https://data-api.polymarket.com/activity
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'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.polymarket.com/activity?user=0xde7be6d489bce070a959e0cb813128ae659b5f4b&type=TRADE&start=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500

```

tart=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500
[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0x99a75dee1487879c95b3d61c92e4493e2c63fc82', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.poly
market.com/activity?user=0x99a75dee1487879c95b3d61c92e4493e2c63fc82&type=TRADE&s
tart=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500
[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0xe72bb501df5306c75c89383d48a1e81073fbb0a0', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.poly
market.com/activity?user=0xe72bb501df5306c75c89383d48a1e81073fbb0a0&type=TRADE&s
tart=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500
[warn] GET failed after retries: https://data-api.polymarket.com/activity
params={'user': '0xb91aeb5accc33a5f9a8615b8ed6b2d352e913987', 'type': 'TRADE',
'start': 0, 'sortBy': 'TIMESTAMP', 'sortDirection': 'ASC', 'limit': 500,
'offset': 3500} err=400 Client Error: Bad Request for url: https://data-api.poly
market.com/activity?user=0xb91aeb5accc33a5f9a8615b8ed6b2d352e913987&type=TRADE&s
tart=0&sortBy=TIMESTAMP&sortDirection=ASC&limit=500&offset=3500
reconstructed flow for 7333 (wallet, token) positions

```

1.3 3. Simulate the lagged copy

Enter at the price `COPY_LAG_HOURS` after their buy; exit at the price `COPY_LAG_HOURS` after their last sell (if they sold half the position) — otherwise hold to resolution. Slippage is paid on both legs.

```

[4]: LAG = CFG.COPY_LAG_HOURS * 3600
S = CFG.SLIPPAGE_TOLERANCE

def simulate_row(r):
    asset = r["asset"]
    buy = price_at(asset, r["entry_ts"] + LAG)
    if buy is None:
        return None
    buy = buy * (1 + S) # we pay up entering
    ↪late
    if not (CFG.PRICE_FLOOR <= buy <= CFG.PRICE_CEILING):
        return None
    f = flow.get((r["wallet"], asset), {"buy": 0.0, "sell": 0.0, "last_sell_ts":
    ↪ 0})
    sold_enough = f["sell"] >= 0.5 * max(f["buy"], 1e-9) and f["last_sell_ts"]
    ↪ r["entry_ts"]
    if sold_enough:
        sell = price_at(asset, f["last_sell_ts"] + LAG)
        if sell is None:
            sell = r["won"] # fallback:
    ↪resolution value

```

```

        else:
            sell = sell * (1 - S)                                # we receive less
    ↪ exiting late
        else:
            sell = float(r["won"])                                # held to resolution
    ↪ (1 if won else 0)
        return (sell - buy) / buy                                # copier's return on
    ↪ the position

sample["copy_ret"] = sample.apply(simulate_row, axis=1)
sim = sample.dropna(subset=["copy_ret"])
print(f"{len(sim)} positions simulated (rest skipped: no price data or out of
    ↪ band)")

```

104 positions simulated (rest skipped: no price data or out of band)

1.4 4. Verdict — lagged mirror vs the frictionless ceiling

```

[5]: def summ(df, n):
    s = df[df["backers"] >= n]
    if s.empty:
        return None
    return {"min_backers": n, "n": len(s),
            "lagged_mean": round(s["copy_ret"].mean(), 3),
            "lagged_median": round(s["copy_ret"].median(), 3),
            "%_profitable": round((s["copy_ret"] > 0).mean(), 3)}

res = pd.DataFrame([r for r in (summ(sim, n) for n in [1, 2, 3]) if r])
print(f"LAGGED MIRROR ({CFG.COPY_LAG_HOURS}h delay, slippage both legs),
    ↪ out-of-sample:")
print("(compare to nb08 ceiling: mirror_median +0.057 / +0.163 / +0.199)")
res

```

LAGGED MIRROR (6h delay, slippage both legs), out-of-sample:
 (compare to nb08 ceiling: mirror_median +0.057 / +0.163 / +0.199)

```

[5]:   min_backers    n  lagged_mean  lagged_median  %_profitable
0           1  104         0.374         0.183         0.625
1           2   33         0.551         0.158         0.727
2           3   13         0.582         0.130         0.615

```

1.5 5. The decision

- **lagged_median** clearly > 0 on a healthy $n \rightarrow$ the exit edge survives realistic delay. This is a genuine, tradeable copy strategy. Next steps: re-check across `COPY_LAG_HOURS` (1h / 6h / 24h) for robustness, then **paper-trade the live mirror** for 4–6 weeks before any capital, and only then consider gated execution (Blueprint §11).

- **lagged_median 0 or negative** → the ceiling was real but only reachable with zero latency; being a step behind erases it. That is the wall, and it’s an honest, well-earned place to stop.

1.5.1 Read it carefully

- Try a few values of `CFG.COPY_LAG_HOURS`. If the edge only exists at 1h but dies by 6–24h, it demands near-real-time infrastructure, which changes the project entirely.
- Survivorship (today’s leaderboard) and equal-weighting still apply. And this is a *sample* of positions — raise `CFG.MIRROR_SAMPLE` to tighten the estimate once you see it work.
- The exit rule (“sold half → exit at last sell”) is a reasonable approximation of messy, partial real-world selling; a positive result would justify modelling it more precisely.